

Your grade: 100%

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1. Given two sets A and B, which of the following is equivalent to  $A \cup B$ ? 1 / 1 point
- ☒  $B \cup A$

☐  $B \cap A$

☐  $B - A$

☐ None of these answers

✔ Nice work

The union of sets is commutative, hence,  $A \cup B = B \cup A$

2. Let A and B be two sets of the universal set  $U$ . Which of the following is equivalent to  $A \cap (B \cap \overline{A})$ ? 1 / 1 point
- ☒  $\emptyset$

☐  $A \cap B$

☐  $U$

☐ None of these answers

✔ Nice work

The intersection of sets is commutative and associative . Hence

$$A \cap (B \cap \overline{A}) = A \cap B \cap \overline{A} = B \cap A \cap \overline{A} = B \cap \emptyset = \emptyset$$

3. Let A , B and C be three sets of the universal set  $U$ . Which of the following is equivalent to  $(A \cup B) \cap C$  1 / 1 point
- ☒  $(A \cap C) \cup (B \cap C)$

☐  $A \cup (B \cap C)$

☐  $(A \cup C) \cap (A \cup C)$

☐ None of the above

✔ Nice work

The intersection of sets is distributive over the union of sets.

4. Let A and B be two sets. Which of the following is equivalent to  $\overline{\overline{A \cup B \cup A}}$  1 / 1 point
- ☒  $A$

☐  $\overline{A} \cap \overline{B}$

☐  $\overline{A}$

☐  $\overline{A} \cup \overline{B}$

✔ Nice work

$$\overline{\overline{A \cup B \cup A}} = \overline{(\overline{A \cap B}) \cup \overline{A}} = (\overline{A} \cup \overline{B}) \cap \overline{\overline{A}} = (A \cup B) \cap A = A$$

5. Let A, B and C be three sets. Which of the following is equivalent to  $\overline{(A \cup B \cup C)} \cap B$ ? 1 / 1 point
- ☐  $\overline{A} \cap \overline{B} \cap \overline{C}$

☒  $\emptyset$

☐  $A$

☐  $B$

✔ Nice work

$$\overline{(A \cup B \cup C)} \cap B = \overline{A} \cap \overline{B} \cap \overline{C} \cap B = \overline{A} \cap \overline{C} \cap \overline{B} \cap B = \overline{A} \cap \overline{C} \cap \emptyset = \emptyset$$